

Prepare — Fujitsu, Software Engineer Apprentice (Job ID 11840)

TAG: Fujitsu - 11840

Generalist entry-level apprenticeship JD (Fujitsu's 1Finity optical networking / infrastructure team). It wants CS fundamentals and systems basics over any specific framework. Your core strengths (DS&A, C++ via CodeChef, full-stack shipping experience) already carry the required bar — the gaps below are genuinely new material, not things to fake.

MUST-priority gaps

1. Computer Networking fundamentals (TCP/IP, HTTP/HTTPS)

- **What it is:** the OSI/TCP-IP layer model, how a packet moves from browser to server, DNS resolution, the HTTP request/response cycle, status codes, headers, TLS handshake basics.
- **Why the JD wants it:** this role touches optical networking / infrastructure management — even at apprentice level you need to reason about the network stack, not just call `fetch()`.
- **The 20% to learn:** you already use this daily without naming it — every API call in Echo/Vibe/Clink AI is HTTP over TCP. Learn: TCP 3-way handshake (SYN, SYN-ACK, ACK) vs UDP; HTTPS = HTTP wrapped in TLS; status code families (2xx/3xx/4xx/5xx). 30–45 min: read a "how does typing a URL work" explainer end to end.
- **Priority: MUST.**

2. Linux environment fluency (beyond "I use it")

- **What it is:** JD lists Linux as required, not nice-to-have — shell comfort: navigation, permissions, processes, scripting, package management.
- **Why:** backend/infra work likely runs on Linux servers; debugging happens over SSH.
- **The 20%:** you're already a daily Fedora user — the gap, if any, is production-server basics: `systemctl`, `journalctl/tail -f`, `chmod/chown`, basic bash scripting.
- **Priority: MUST** (per JD), likely already close — verify in interview prep rather than deep-study.

GOOD-priority gaps

3. Cloud platforms — Azure / GCP specifically

- JD lists AWS/Azure/GCP as alternatives — you have genuine AWS experience (Echo uses AWS Secrets Manager); Azure/GCP unlearned.
- **The 20%:** AWS already satisfies "cloud platforms." If time allows, skim Azure equivalents (Blob Storage ~ S3, Functions ~ Lambda, DevOps ~ GitHub Actions).
- **Priority: GOOD** — don't over-invest.

4. Java (as a third "or" language option)

- JD says "Python, C++, Java, or similar" — Python + C++ already satisfy this on their own. Deliberately left off this resume variant.
- **Priority: SKIP for this JD.**

Talking points for this interview

- **DS&A depth:** lead with 6-star CodeChef (2200+, peak 2212) and 1000+ DSA problems (75% Medium/Hard) — your strongest, most literal match to "programming fundamentals and data structures."
- **C++ fluency:** all competitive programming is in C++ — be ready to discuss complexity analysis, STL containers, and a recent hard problem.
- **SDLC / Agile exposure:** Agile/Scrum at Clink AI (Jira) — connect directly to "apply software engineering best practices and Agile methodologies."

- **Docker, real usage:** Vibe's E2B sandbox is built from a real custom Dockerfile — concrete, verifiable Docker experience.